# Mohit Dubey

**AI Full Stack Engineer**

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## SUMMARY

Senior Full Stack / AI Engineer with 5+ years building production **Python** backend services and **Generative AI / Agentic AI** systems end to end: multi-agent orchestration, planning, tool-calling, and memory/state management (**LangChain**, **LangGraph**, **DeepAgents**). Hands-on with **RAG** architectures (embeddings, vector databases, chunking, retrieval optimization), **MCP** tool-integration standards, low-code/configuration-driven agent platforms for business users, and **FastAPI** and **asyncio** services with SSE/WebSocket streaming, resilience patterns, and structured observability. Experienced with **Kubernetes** and **OpenShift**, **MongoDB** for state/checkpointing, **CI/CD** (**GitHub Actions**, **Jenkins**, SonarQube), and enterprise auth (AD/SSO, RBAC, OAuth2).

## WORK EXPERIENCE

**Software Engineer —— Publicis Sapient** June 2025 - Present

- Led architecture of an enterprise-grade **GraphRAG**-based private coding assistant using **LangGraph**, **DeepAgents**, and agentic AI, improving retrieval accuracy by 38% and cutting developer resolution time by 60%.
- Designed a **low-code, configuration-driven agent platform** letting non-technical business users configure, deploy, and manage AI agents from reusable templates for automation, analytics, and predictive use cases.
- Built an **MCP server** with dynamic tool discovery, semantic search, and context orchestration to connect agents with enterprise systems/APIs, cutting LLM context overhead by 70% and lowering inference cost.
- Integrated **LiteLLM** as a unified internal/third-party LLM gateway with cost-based fallback, rate limiting, and centralized usage tracking across agent workloads.
- Orchestrated **multi-agent** planning, tool-calling, and memory across specialized agents with **MongoDB**-backed state and checkpointing, **SQL Server** and **Oracle** integrations for enterprise data access, guardrails, and policy enforcement for secure multi-tenant deployment.
- Architected **RAG** pipelines on **LangGraph** — embeddings, vector databases, chunking strategies, and hybrid retrieval/reranking — increasing answer relevance by 42% for thousands of concurrent enterprise requests.
- Exposed agent capabilities via **FastAPI**, **Flask**, and **asyncio** with SSE/WebSocket streaming responses, circuit-breaker/retry resilience patterns, and structured logging for production observability.
- Instrumented agents with **Arize** (eval, drift, tracing) and **Datadog** (APM); deployed on **Kubernetes** and **OpenShift** via **GitHub Actions** and **Jenkins** CI/CD with SonarQube code-quality gates.
- Secured platform access with **AD/SSO** authentication and **RBAC**, and **OAuth2** for external-facing workflows; partnered with execs on AI opportunity/ROI and go-to-market transformation.
- Mentored engineers on agentic system design, prompt engineering, and AI safety/guardrails; owned delivery lifecycle end-to-end, maintaining a 95% project success rate.
- Designed and implemented a **client-server UI version synchronization system** (React + Nginx) to ensure users always run the latest deployed build.
- Built a **reusable, highly configurable table component** adopted across 15+ modules, improving consistency and reducing maintenance overhead.
- Centralized API interactions by creating a **shared network layer** with consistent error handling and retry logic, improving reliability across the application.
- Implemented **performance-optimization** techniques such as **code-splitting, lazy loading, and dynamic imports** to reduce **JavaScript bundle size** and improve time to interactive (TTI).
- Engineered scalable **Frontend Architecture** using **React.js**, **Next.js**, and micro-frontends, optimizing platform performance by **60%** and securing workflows using **OAuth2** authentication protocols.

**Software Engineer —— Deqode** Aug 2023 - May 2025

- Architected enterprise-grade **RAG pipeline** using **LangChain**, Pinecone/FAISS, and AWS, reducing LLM hallucinations by 32% and improving retrieval latency by 45% across 1M+ documents.
- Fine-tuned 7B/13B **Qwen3-VL** LLMs using **SFT**/**QLoRA** on **AWS SageMaker** GPU clusters, improving domain accuracy by 21% while cutting inference cost by 60%.
- Deployed production **LLM serving** on **GCP Vertex AI** with **vLLM**/TensorRT-LLM, serving 13B-parameter models at sub-200ms latency for 500K+ requests/day.
- Orchestrated cloud-native ML systems on **Kubernetes** with **MLflow** and Terraform; led platform modernization for 15+ ML teams, compressing deployment cycle from 3 weeks to 2 days.
- Deployed multi-agent conversational systems using LangChain Agents and AWS Bedrock API gateways; **mentored engineers** on RAG scaling and context-window optimization.
- Built **AI monitoring/observability dashboards** (Grafana, Power BI, Looker) tracking operational metrics, system health, and model drift.
- Built a **configuration-driven, schema-to-React renderer** supporting dynamic layouts, conditional fields, and versioned JSON configs for non-engineering UI rollout.
- Created robust **AI monitoring and real-time dashboarding infrastructures** using Grafana, Power BI, and Looker to continuously track operational metrics, system health, and model drift.

- Translated deep AI behavioral insights into predictive analytics models, enhancing customer experience metrics and optimizing down-stream system performance.
- Built Aadhaar-Voter ID linking UI in React using **Bootstrap** for responsive layouts, scaling to **handle high-traffic production loads** with resilient, reusable components.
- Migrated **Redux** to modular **RTK** slices + RTK Query, reducing API overfetching by 78%.
- Developed editable virtualized grid (1M+ rows) using **react-window**, with smooth scroll and memoization.
- Led federated design **system adopted by 50+ teams**, enforced standards.
- Integrated Web Vitals + **Sentry** to track LCP, FCP, TTI across 500k+ sessions/month and optimize core web metrics.
- Built full-stack applications with **React**, **TypeScript**, and **NestJS**, shipping REST API backends backed by **PostgreSQL**, **Oracle**, and **MongoDB** serving 100K+ users with sub-100ms p99 latency via Redis caching.

**Software Engineer —— Shiv Technolabs** Feb 2023 - July 2023

- **Leveraged deep knowledge of browser internals** to optimize performance, systematically reducing reflow,repaint, and bundle sizes to achieve significant improvements in Core Web Vitals.
- **Implemented advanced state management** patterns across complex application workflows using MobX and modern ES6+ JavaScript features for predictable data flow.
- **Led frontend performance audits** analyzing JS engines compilation behaviors and execution contexts to eliminate memory leaks and runtime bottlenecks.
- Orchestrated the transition to **Next.js** methodologies, integrating **RESTful APIs** to achieve seamless data transfer across platforms, which decreased website bounce rate by **15%** and grew traffic.

**Software Engineer —— Sourcebae** May 2021 - Jan 2023

- Built full-stack applications with **React**, **TypeScript**, and **NestJS**, shipping REST API backends backed by **PostgreSQL** and **MongoDB** serving 100K+ users with sub-100ms p99 latency via Redis caching.
- Designed a reusable **TypeScript** component library with centralized state management, cutting feature delivery time by 40% across 5 concurrent product teams.
- Containerized services with **Docker** and established **CI/CD pipelines** with automated testing and staged rollouts, eliminating manual deployment errors and enabling zero-downtime releases.
- Built **offline-first UI** in React Native using SQLite for local data storage and background sync to update the server when reconnected.

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## PROJECTS

**TradeBrains — Automated Real-Time Algorithmic Trading Platform**

- Architected an end-to-end algorithmic trading platform predicting 20-minute stock movements by fine-tuning open-source generative models (**Llama 3.3 70B**) using 3-stage **RLHF** (**SFT**, **Reward Model**, **PPO**) and **GRPO** to align prediction outputs directly with target performance metrics.
- Implemented custom **GRPO/PPO reward functions** integrating multi-objective metrics (factual correctness, formatting compliance, and financial risk metrics) to eliminate hallucinations in high-stakes domain outputs.
- Engineered **QLoRA 4-bit** parameter-efficient fine-tuning (**PEFT**) pipelines for 8B-parameter sentiment models, reducing memory footprints while maintaining **88% structured score extraction accuracy**.
- Executed **Direct Preference Optimization (DPO)** and **KTO** fine-tuning on domain-specific instruction datasets, bypassing separate reward model latency while matching RLHF alignment quality.

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## EDUCATION

**Rajiv Gandhi Proudyogiki Vishwavidyalaya** Bhopal, India

Bachelor of Engineering (HONOURS) - Information Technology; GPA: 7.75 August 2016 - May 2020

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## ACHIEVEMENTS

- **AWS Certified Developer Associate**.
- **AWS Certified AI Practitioner**.
- **AWS Certified Cloud Practitioner**.
- **Hacker Rank Javascript Basic**.
- **Hacker Rank CSS certification**.
- **Sololearn React + Redux**

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## TECHNICAL SKILLS

- **LLMs & Agents**: LangChain, LangGraph, CrewAI, Google ADK, LlamaIndex, Hugging Face Transformers, OpenAI API, Ollama, vLLM, TensorRT-LLM, LiteLLM, MCP Servers, Agent Orchestration, RAG Systems, Prompt Engineering
- **Training & Optimization**: LLM Fine-Tuning (SFT, QLoRA, LoRA, PEFT), RLHF, GRPO, DeepSpeed, FSDP, Distributed Training, CUDA, Model Quantization (INT8, NF4)
- **ML & Deep Learning**: PyTorch, TensorFlow, scikit-learn, XGBoost, Classical ML, Feature Engineering, Model Optimization
- **Data Engineering**: Apache Spark (PySpark, Spark SQL, Scala), Databricks (Unity Catalog), Snowflake, Delta Lake, Kafka, Airflow, BigQuery, ETL/ELT, Data Modeling (Dimensional, Relational), Feature Store (Feast)
- **MLOps & Infrastructure**: MLflow, Kubeflow, Kubernetes, OpenShift, Docker, Terraform, CI/CD (GitHub Actions, Jenkins, SonarQube), Model Monitoring & LLM Observability (Arize, Datadog), Drift Detection, A/B Testing

- **Cloud**: AWS (SageMaker, EKS, Lambda, S3), GCP (Vertex AI, GKE, Dataproc, BigQuery), Azure
- **Backend Engineering**: Python, Scala, FastAPI, Flask, asyncio, SSE/WebSockets, Go, TypeScript, REST APIs, gRPC, Microservices, Redis, n8n
- **Databases & Vector Stores**: PostgreSQL, MongoDB, SQL Server, Oracle, Redis, Qdrant, Milvus, Weaviate, Chroma, SQL
- **Security & Auth**: OAuth2, AD/SSO, RBAC
- **Frameworks:** Reactjs, Redux, Nextjs, SCSS, MaterialUI, ExpressJs, Jest, RTL, Enzyme
- **Tools:** Git/Github, Postman, Swagger, Figma, Sequelize ORM, Firebase, Nginx, Webpack, Babel, ESLint, Prettier
- **Other:** RESTful Web Services, Agile methodologies, Unit testing, Integration testing, micro frontends, Incident Response & Operational Excellence